# Centripetal Force: Weighing a Stopper With a Stopwatch

**SYNOPSIS**

A rubber stopper on a string is whirled in a horizontal circle above your head. The string runs down through a plastic tube, and hanging from its lower end is a stack of metal washers. Those washers are not decoration: their weight *is* the force holding the stopper in its circle. So you have a system where the centripetal force is something you can set by hand — add four more washers and you have added a known number of newtons — and the speed is something you can measure with a stopwatch. Spin faster and the required force rises. The question is whether it rises the way $F = mv^2/r$ says it should, and the honest way to answer that is not to check one run but to take five, plot them, and look at the slope of the line they fall on.

**Due** see the Module 5 Lab folder on Brightspace | **Report** Full lab report — use the Lab Template | **Pairs with** M5 N1 | **Data** provided — this is a virtual lab

**PERFORMANCE EXPECTATIONS** what you should be able to do when this lab is written up

- **PE1** **Explain why the weight of the hanging washers equals the centripetal force on the stopper**, by treating the string as one object and the hanging stack as another, and saying what makes the tension the same at both ends.
- **PE2** **Convert a count of revolutions and a total time into a speed**, through the period and the circumference, and carry the units at every step.
- **PE3** **Rearrange $F_c = mv^2/r$ into the equation of a straight line** — $F_c = (m/r)\,v^2$ — and say what the slope must physically be if the model is right.
- **PE4** **Obtain a mass from the slope of a graph**, with its uncertainty, and compare it against the mass read off a balance.
- **PE5** **Distinguish a slope result from a single-point result**, and explain why a non-zero intercept wrecks the second while leaving the first intact.
- **PE6** **Identify the direction of a systematic offset from the sign of the intercept**, and propose a mechanism in the apparatus that would produce it.

## Where this sits against the Module 5 notes

**M5 N1 gives you the kinematics; this lab supplies the dynamics.** Period, frequency, angular velocity and the relation $v = \omega r$ are all from that note, and you will use every one of them in Part II. What the note does not yet do is connect that motion to a *force* — that is what this lab is for, and the argument you build here is the one the centripetal-force note will later formalise.

Where each part of the note belongs in this report. The rubric awards points for these.

| Report section | What you bring from M5 N1 |
|---|---|

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|---|---|
| **Introduction** | The definition of centripetal force and the statement $F_c = mv^2/r$, with every symbol defined, plus **your own rearrangement** into the straight-line form. End with a hypothesis you can later confirm or reject. |
| **Experimental** | The labelled setup drawing, redrawn by you, showing the stopper, the tube, the string, the paper-clip marker and the washers. State what you are holding constant and what you are varying. |
| **Analysis** | The chain from a stopwatch reading to a speed: 30 revolutions → period → circumference → v → $v^2$. Then the graph and its slope. |
| **Discussion** | Whether the slope recovers the stopper's mass, what the intercept says, and why the two methods of getting m disagree by as much as they do. |

Read or reprint the note here: Module 5 Note 1 — Circular Motion Kinematics.

### The physics you are using

Two statements, and the whole lab is the place where they meet.

**One.** An object moving in a circle of radius r at speed v is accelerating toward the centre, and the net force producing that acceleration is

$$F_c = m_{stopper}\, v^2 / r$$

**Two.** The string runs over the lip of the tube and down to the washers. If the washers hang without moving up or down, the net force on them is zero, so the tension in the string equals their weight:

$$T = M_{hanging}\, g$$

That tension is transmitted along the string, and at the top it is what pulls the stopper inward. So $\mathbf{T = F_c}$, and the force on the stopper is a quantity you set by counting washers rather than one you have to measure. Putting the two together and rearranging into the form of a straight line:

$$F_c = ( m_{stopper} / r ) \times v^2$$

Plot $F_c$ up the page against $v^2$ across it and, if the model holds, you get a line through the origin whose **slope is $m_{stopper}/r$**. Both of those you know independently — the stopper went on a balance and the radius was set with a paper clip — so the graph makes a prediction you can check.

> **What is held constant, and what is not.** The radius is fixed at 50.0 cm by a paper clip taped below the tube, and the stopper is the same stopper all the way through. The only thing that changes between runs is the number of washers, and therefore the force. Your Experimental section must say this explicitly — if the radius drifted while you spun faster, the slope would be measuring a moving target and would mean nothing.

> **Predict first. You double the number of hanging washers, which doubles the centripetal force. What happens to the speed of the stopper?**

(A) It doubles.

(B) It rises by a factor of √2, about 1.41.

(C) It rises by a factor of 4.

(D) It stays the same — the stopper's mass has not changed.

**Why that question is worth asking before you start.** The force depends on the *square* of the speed, so the speed depends on the square root of the force. Doubling the washers buys you only about a 41% increase in speed, which is exactly why the runs in this data set are bunched together at the fast end: going from 4 washers to 20 — five times the force — raises the speed by a factor of only √5 ≈ 2.24. It also tells you something about the graph: a plot of $F_c$ against $v$ would be a curve and hard to judge by eye, while a plot against $\mathbf{v^2}$ is a straight line. Choosing what to put on the horizontal axis is part of the analysis, not a formatting decision.

## The apparatus

A size-4 rubber stopper is tied to one end of 1.5 m of nylon string. The string is threaded down through a plastic tube 10–20 cm long, and a bent paper clip on the free end carries the metal washers. A second paper clip is taped to the string just below the tube to mark the radius: the stopper is whirled at whatever speed keeps that marker hovering just below the tube without touching it, which is what holds the radius fixed at 50.0 cm.

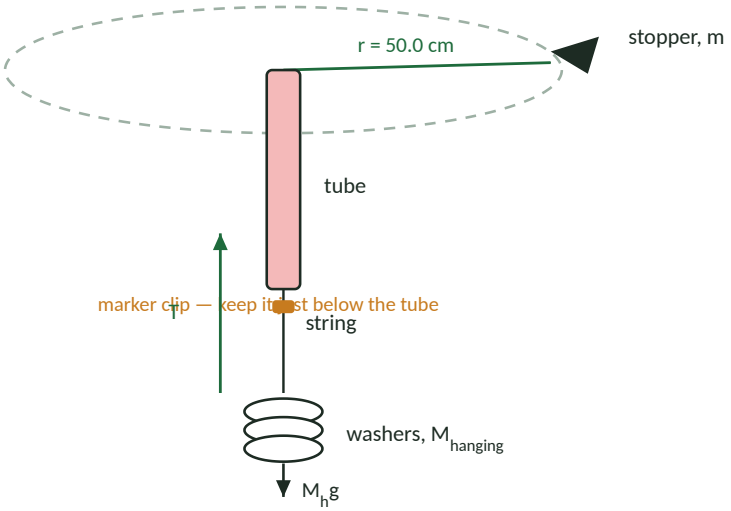



Figure 1. The apparatus as the data were taken. The radius is fixed by the marker clip; the hanging washers set the tension, and that tension is the centripetal force on the stopper.

### The constants

| Quantity | Measured value | In SI |
|---|---|---|
| 10 washers | 70.07 g | 7.007 g each |
| Rubber stopper, m | 10.09 g | 0.01009 kg |
| Paper clip (on the hanger) | 1.60 g | 0.00160 kg |
| Radius, r | 50.0 cm | 0.500 m |

**The paper clip counts.** It hangs from the string along with the washers, so it is part of $M_{hanging}$ in every run. Leaving it out shifts every force by 0.0157 N — small, but it is a *systematic* shift in one direction, which is exactly the kind of thing that moves an intercept. Include it and say that you did.

## Part I — The data

Five runs, provided in the workbook. In each, the stopper was whirled at the speed that kept the marker clip just below the tube, and a partner timed **30 complete revolutions**.

The five runs as recorded. Everything else in this lab is derived from these two columns.

| Washers | $M_{hanging}$ incl. clip (kg) | Time for 30 rev (s) |
|---|---|---|
| 4 | 0.02963 | 20.86 |
| 8 | 0.05766 | 17.42 |
| 12 | 0.08568 | 14.72 |
| 16 | 0.11370 | 12.34 |
| 20 | 0.14170 | 11.29 |

## Part II — From a stopwatch reading to a speed

Four steps per run, and each one is a sample calculation the rubric asks to see worked once.

I. **The centripetal force.** $F_c = T = M_{hanging}\, g$, with $g = 9.8$ m/s². This is the only place g enters the lab.

II. **The period.** Divide the total time by 30 revolutions to get the time for one.

III. **The speed.** The stopper covers one circumference $2\pi r$ in one period, so $v = 2\pi r / P$. Equivalently $\omega = 2\pi/P$ and $v = \omega r$ — use whichever route you prefer, but say which.

IV. **Square it.** $v^2$ is what goes on the horizontal axis.

**Why 30 revolutions and not one.** A human thumb on a stopwatch is good to something like a tenth of a second. One revolution here takes about half a second, so timing a single one would carry a 20% error. Timing thirty and dividing spreads that same tenth of a second across thirty periods and brings the error down to well under a percent. This is a general technique and the Discussion is a good place to say so.

## Part III — The graph that weighs the stopper

Plot **$F_c$ on the vertical axis against $v^2$ on the horizontal**, one point per run, and fit a straight line. From the model,

$$\text{slope} = m_{stopper} / r \quad \rightarrow \quad m_{stopper} = \text{slope} \times r$$

so the line hands you a mass you never put on a balance. Compare it against the 10.09 g the balance gave, and quote a percent difference.

### Fit it twice, and report both

Fit the line **once with an intercept and once forced through the origin**, and report the mass each gives. The model says the intercept should be zero — no speed, no force — so the two fits ought to agree. Whether they do is a result in its own right, and the gap between them is the most informative number in this lab.

### What counts as agreement

A percent difference of **10% or less** between the mass from the slope and the mass from the balance is a confirmation. If you are outside that, check in this order: the paper clip left out of $M_{hanging}$; r entered in centimetres; the period taken as the raw 30-revolution time; or $v^2$ and $F_c$ swapped between the axes.

## Check your analysis

Paste your five force-and-speed-squared pairs to check the line you fitted. Your report still has to show the graph and a worked sample calculation of each type, by hand, with units — this tool checks arithmetic, it does not replace the analysis.

**TOOL · FIT $F_c$ AGAINST $V^2$, AND READ OFF THE MASS**

**One run per line: $v^2$ in $m^2/s^2$, then $F_c$ in N, separated by a comma, a tab or a space**

**Stopper mass from the balance (g)**

10.09

**Radius r (m)**

0.500

Uncertainties are the standard least-squares errors on the fitted parameters. Both fits are computed every time, because the comparison between them is the point.

## Additional Questions

**Choose three of the five below and answer those three fully.** Name which three you chose. Answer them in a section of their own headed **Additional Questions**, in **sentences and paragraphs** with the working shown, never as a bare numbered list and never by restating the question before answering it. Depth on three beats coverage of five; this section is worth 8 points on the rubric.

**AQ1 — the intercept, and what it is telling you.** Your fitted line does not pass through the origin, and the model says it should. Report the intercept with its sign and its units. Then work out what it would take to produce one: a force that acts on the stopper regardless of how fast it is going, or a force in the tension that never reaches the stopper. Propose one concrete mechanism in this apparatus, estimate its size from the intercept you measured, and say whether that estimate is physically reasonable. Finish by stating which of your two mass values — free fit or origin-forced — you would report as *the* result, and defend the choice.

**AQ2 — why one run is not enough.** Take your fastest run alone and compute the stopper's mass directly from $m = F_c r/v^2$. Do the same for your slowest run. The two answers will differ substantially, and both will differ from the slope result. Explain, in terms of the intercept, why a single point is so much worse than a line through five — and why the *slowest* run is the worst of all. Generalise it: state the condition under which taking a single measurement and dividing is a legitimate way to get a constant, and the condition under which it is not.

**AQ3 — the string is not horizontal.** The treatment above assumes the stopper moves in a horizontal circle with the string lying flat. It cannot: the string must tilt downward from the stopper to the tube or there would be nothing holding the stopper up against gravity. Draw the free-body diagram for the stopper with the string at angle θ below horizontal, resolve the tension into components, and show that the horizontal component supplies the centripetal force while the vertical component supports the weight. Then derive the true relation between T and $v^2$, and estimate θ for your fastest and slowest runs. Does the tilt get better or worse as the stopper speeds up, and does that account for the direction of your intercept?

**AQ4 — propagating the stopwatch.** Suppose the timing of 30 revolutions is good to ±0.2 s. Propagate that into the period, into v, and into $v^2$, and give the resulting uncertainty on $v^2$ for your fastest and slowest runs. (Note that the fractional uncertainty in $v^2$ is twice that in v, and show why.) Compare the size of those horizontal error bars against the scatter of your points about the fitted line. Is the stopwatch the limiting source of error in this experiment, or is something else?

**AQ5 — designing the experiment you were not given.** This lab varies the force and holds the radius fixed. Design the complementary experiment: hold the force fixed and vary the radius instead, which is what Trevor's handout gestures at when it asks what you would change to find the relationship between v and r. State what you would plot against what to get a straight line, what its slope would physically be, and what practical difficulty the apparatus puts in your way that the force-varying version does not have. Predict, from the five runs you do have, roughly what speed a 0.25 m radius would require at 12 washers.

## Prompts for your Discussion

These are **not** the Additional Questions. Work them into your **Discussion** section as part of your argument, in your own order. Do not answer them as a list, and do not limit yourself to them.

I. You have two masses for the same stopper — one from the balance, one from the graph. Which do you trust more, and what would have to be true for the graph to be the better of the two?

II. The model predicts a line through the origin. Yours is not. Is that a failure of the model, a failure of the apparatus, or a failure of the analysis? Argue for one.

III. Your five points span a factor of five in force but only a factor of about 2.2 in speed. Does that spread help or hurt the slope determination, and would you have chosen different washer counts?

IV. What would happen to the graph if the marker clip drifted downward during the fast runs, so the radius quietly grew? Which way would the points move, and would you be able to tell?

V. The lab asks you to keep the radius constant by watching a paper clip. Estimate how well a person can really do that, and fold your estimate into your error discussion.

VI. Trevor's original version of this lab also asks for a plot of v against $F_c$. Why is that graph harder to judge than the one you made, even though it contains exactly the same information?

## Files for this lab

HANDOUT

**This lab, as a PDF**

The whole page, printable. The calculator only works on the web version.

REQUIRED

**Example workbook**

The five runs, a blank analysis sheet laid out for you, and one worked column.

GRADING

**Lab 06 rubric**

Read it before you write, not after.

SUPPORT

**Report starter**

Section guidance only — the sentence stems have gone. See the note inside.

NOTE

**M5 N1 — Circular Motion Kinematics**

Period, angular velocity and v = ωr, which Part II runs on.

REFERENCE

**Formatting data tables**

Table captions, units in headers, significant figures.

## References to get you started

Your references should include material on centripetal force and circular motion. Your textbook, this handout and the Module 5 note pages all count, and you need at least one source beyond the textbook.

1. [Author's names], *Journal or Book title*, edition, Volume number, page number, (year).

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